

CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet 10

Name (Print): _____ Section: _____

Access & Scope

1. Null vs empty

```
int[] [] a = null;
```

```
int[] [] a = new int[0][0];
```

null means: _____

empty array means: _____

2. Complete checks

Type	Check for null	Check for empty	Return an empty instance
String	s == null	s._____	return
Array	arr == null	arr._____ == 0	return
ArrayList	list == null	list._____()	return
HashMap	map == null	map._____()	return

3. Trace it (what happens and why?). For each line, write: **OK**, **compile error**, or **run-time error**, and name the error if applicable (e.g., syntax, null pointer, index out of bounds).

	Result	Why / Error name
int[] [] arr1 = null;	_____	_____
int[] [] arr2 = new int[] [];	_____	_____
int[] [] arr3 = new int[0][0];	_____	_____
arr3[0][0] = 3;	_____	_____
System.out.println(arr1.length);	_____	_____
System.out.println(arr3.length);	_____	_____

4. Null + Methods. If a variable is **null**, we CAN / CANNOT call a method on it.

5. **Review - Size of Objects.** Write the correct way to get the size/length:

```

1 String s = "Hello";           s. _____
2
3 int[] nums = {1,2,3,4,5};      nums. _____
4
5 ArrayList<String> list = new ArrayList<>();  list. _____
6
7 HashMap<String,Integer> map = new HashMap<>(); map. _____

```

6. **Access Modifier Order.** From most restrictive to least restrictive:

7. Identify the access modifier

- 1) _____: available to any other class in the same package
- 2) _____: can only be accessed within the declared class itself
- 3) _____: accessible in the same package and by subclasses in other pkg
- 4) _____: can be accessed from any class (=everywhere)

8. **Access Check (What can you access?)**

```

1 class Player {
2     private int health = 100;
3     int level = 1;
4     protected int score = 50;
5     public int coins = 10;
6 }

```

A) **Inside Player class.** Which fields can be accessed?

B) **Different class, same package.** Which fields can be accessed?

C) **Subclass in a different package.** Which fields can be accessed?

D) Different package, NOT a subclass. Which fields can be accessed?

E) Which modifier is best for encapsulating health?

9. Scope

- 1) _____: anywhere in the object
- 2) _____: anywhere in the class
- 3) _____: within the method only
- 4) _____: within the block

10. Scope in Action (What happens?)

```
1 public class Player {  
2     int health = 100;  
3  
4     public void setHealth(int health) {  
5         health = health;  
6     }  
7 }
```

```
1 Player p = new Player();  
2 p.setHealth(50);  
3 System.out.println(p.health);
```

What is printed? _____

Fix the code:

```
1     public void setHealth(int health) {  
2         health = health;
```

11. Before You Leave

Write one question you still have about today's topic.
